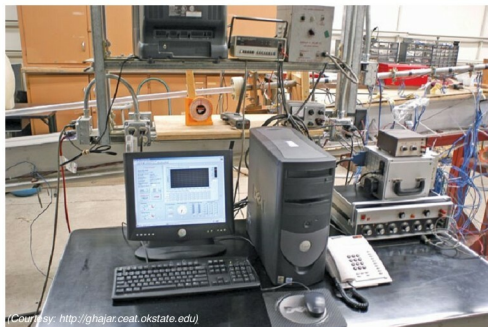


What is New in Data Acquisition Systems

From research work to modern engineering processes, high rates of encoded information in multiple signals from simultaneous processes have forced data acquisition (DAQ) systems to evolve. This article gives you insight on the latest in DAQ systems



(Courtesy: <http://ghajar.coat.okstate.edu>)

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Data acquisition is a process of acquiring information and documenting it in a meaningful way, to analyse a process or phenomenon. Data acquisition (DAQ) systems are usually used in applications that require data logging, control, test automation, monitoring and prototyping. Advances in electronics and information processing technologies have made these types of processes simpler, accurate, versatile and reliable.

What is happening to data acquisition

The most visible trends can be seen as the effects of transition to PC-based DAQ that started some time ago. Most earlier box instruments had processing built into them. Now, all

that processing is being done inside computers, so the instruments are interfaced to a computer with analysis being done through computer software. Thus we see a more software-defined approach to DAQ, as well as the emergence of high-speed USB-enabled DAQs.

The other key trends are wireless DAQs connecting with user's consumer electronics, which users can associate with using sensors, signal conditioning hardware along with flexible software and extensive DAQ channel counts. Integrated and multifunctional applications for modular PCI-based and USB DAQ systems are also gaining popularity due to their comparatively economical nature.

Let us take a look at some recent major developments in the field of DAQ.

Better interfacing options to keep up with DAQ system requirements

The industry is moving towards a trend of having a combination of modular hardware and flexible software. These newer modular systems have appropriate signal conditioning and analogue-to-digital conversion (ADC), with various interfacing sensors supporting multiple DAQ requirements.

The plug-and-play signal conditioning systems improve accuracy and performance by simplifying sensor connectivity, removing noise and ensuring safety. The traditional approach used to be instruments performing specific DAQ tasks, which is not feasible considering today's requirements.

Chinmay Misra, technical marketing engineer, National Instruments says, "A traditional DAQ system is equipped with a fixed user interface and in-built programmed software, and hence it is not capable of catering to multiple applications. This means that as soon as DAQ requirements change, the traditional set-up has to be completely revamped. With today's advanced technology, as upgrade cycles grow relentlessly shorter and relentlessly more cross-platform, the traditional approach is neither feasible nor cost-effective."

Software-defined, modular instrumentation is gaining popularity as it offers flexibility, scalability and higher speeds. A modular instrument is essentially a generic hardware that acquires and generates raw data while the software is stored on the user's PC,